An unavoidable even-power term in Volterra singular values

Disproof of Conjecture 00000009778

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Abstract

The classical Volterra operator on $L^2(0,1)$ has ordered singular values $s_n = 1/[\pi(n-1/2)]$, $n \geq 1$. Consequently $n^2(s_n - 1/(\pi n)) \to 1/(2\pi) \neq 0$, excluding an expansion with only odd inverse powers. A full unrestricted expansion does exist; its even-power terms cannot be omitted. We give the actual operator-to-spectrum argument and explicitly separate it from the Lean-checked integral identities and asymptotic core. The Hilbert-space spectral identification is a stated mathematical bridge, not a claimed complete Lean formalization.

The actual compact operator

On the real or complex Hilbert space $H = L^2(0,1)$, set

$$(Vf)(x) = \int_0^x f(t)\,dt.$$

Its kernel $\mathbf{1}_{\{0\leq t\leq x\leq 1\}}$ is precisely an interval-indicator integration kernel, as required by the source. Its squared L^2 kernel norm is $1/2$, so V is a bounded Hilbert–Schmidt operator and is compact. Every L^2 function on this finite interval is integrable, and the formula selects an absolutely continuous representative. Kernel diagonal conventions have no effect on the operator on L^2 equivalence classes.

Fubini applied to the inner product gives the actual adjoint

$$(V^*g)(x) = \int_x^1 g(t)\,dt.$$

Thus $A = V^*V$ is compact, positive and self-adjoint, with

$$(Af)(x) = \int_x^1 \int_0^t f(s)\,ds\,dt = \int_0^1 \big(1 - \max(x,s)\big) f(s)\,ds.$$

Fubini is justified, for example, by integrability of $|f(s)||g(t)|$ on the square, using $L^2 \subset L^1$ and Cauchy–Schwarz. Positivity follows from $\langle Af, f\rangle = \|Vf\|_2^2$.

For arbitrary $f \in H$, the displayed representative $y = Af$ satisfies

$$y \in C^1[0,1], \qquad y'(x) = -\int_0^x f(s)\,ds, \qquad y'' = -f \text{ a.e.}, \qquad y'(0) = 0,\ y(1) = 0.$$

Here y' is absolutely continuous; no classical second derivative of an arbitrary L^2 function is assumed.

Complete spectral identification and ordering

Suppose $Af = \lambda f$ in L^2, with $f \neq 0$ and $\lambda \neq 0$. Positivity gives $\lambda > 0$. Replace f by the representative $\widetilde{f} = y/\lambda$. It is continuous, so the integral formula for y' and the fundamental theorem of calculus upgrade y to C^2. Consequently

$$\widetilde{f}'' + \lambda^{-1}\widetilde{f} = 0, \qquad \widetilde{f}'(0) = 0, \quad \widetilde{f}(1) = 0$$

hold classically everywhere. Writing $\omega = 1/\sqrt{\lambda} > 0$, the nonzero solutions are multiples of $\cos(\omega x)$, and $\cos\omega = 0$. Therefore

$$\omega_n = (n - \tfrac{1}{2})\pi, \qquad \lambda_n = \omega_n^{-2}, \qquad n = 1, 2, \ldots.$$

Conversely direct integration proves

$$V\cos(\omega_n x) = \frac{\sin(\omega_n x)}{\omega_n}, \qquad V^*\sin(\omega_n x) = \frac{\cos(\omega_n x)}{\omega_n}.$$

Each listed eigenvalue thus occurs, with a one-dimensional eigenspace by the ODE classification. There are no other nonzero eigenvalues. The compact self-adjoint spectral theorem applies to A, not to the non-self-adjoint operator V. It gives exhaustiveness of the eigenvalue list with multiplicities. Also $\ker A = \ker V = \{0\}$: if $Vf = 0$ a.e., its continuous primitive is zero everywhere, hence its derivative f vanishes a.e. The positive eigenvalues above strictly decrease to zero. Taking their positive square roots gives the ordered singular values

$$s_n(V) = \sqrt{\lambda_n} = \frac{1}{\pi(n - 1/2)}, \qquad n \geq 1.$$

This agrees with the squared Gram spectrum stated in Opmeer [2], Example 31. The positive square root is essential; the eigenvalues of V^*V are not the singular values of V.

The missing even term

Exact algebra yields

$$n^2\left(s_n - \frac{1}{\pi n}\right) = \frac{n}{\pi(2n-1)} \longrightarrow \frac{1}{2\pi} \neq 0.$$

An odd-only Poincaré expansion in n^{-1} would, already through order three, give a real c such that

$$n^3\left(s_n - \frac{1}{\pi n} - \frac{c}{n^3}\right) \longrightarrow 0.$$

Multiplying this limit by $1/n$ and adding c/n would force the previous n^2-scaled difference to tend to zero, a contradiction.

Theorem 1. *The Volterra singular values have a nonzero n^{-2} coefficient, so they do not admit the conjectured odd-only inverse-power expansion. This remains true under every integer change of starting index.*

Proof. The first assertion follows from the nonzero limit on page 2. For an integer shift k, eventually $s_{n+k} = 1/[\pi(n+a)]$ with $a = k - 1/2 \neq 0$. Then

$$n^2 \left(s_{n+k} - \frac{1}{\pi n} \right) = -\frac{a}{\pi} \frac{n}{n+a} \longrightarrow -\frac{a}{\pi} \neq 0.$$

The same necessary odd-expansion condition is impossible. □

The ordinary full expansion actually is geometric:

$$s_n = \frac{1}{\pi n} \left(1 + \frac{1}{2n} + \frac{1}{4n^2} + \cdots \right).$$

Every displayed coefficient is nonzero. If "odd corrections" were instead intended inside the relative factor, the relative n^{-2} term $1/4$ also disproves that reading. A half-integer frequency variable $n - 1/2$ is not an integer reindexing or an expansion in the stated integer n. We refute the parity restriction, not the existence of an unrestricted full expansion; the additional Bernoulli-coefficient clause need not be settled.

Formalization boundary and reproduction

Lean defines genuine forward and backward interval integrals on real functions. It proves their displayed cosine-mode and Gram-mode identities for every mode and every real argument, including the half-integer boundary condition. Separately it proves the nonnegative square-root step from the published Gram formula, the actual scaled-difference limit, and failure of the necessary third-order odd expansion. Theorems quantify over any sequence eventually satisfying the exact spectral formula. They include every integer offset; $n = 0$ is immaterial to the eventual statements.

The actual L^2 bounded operator, adjoint theorem, regularity upgrade, compact spectral theorem and exhaustive ordered spectral identification on pages 1–2 are mathematical analysis outside this Lean project. They are substantive bridges, not definitions of a chosen sequence as singular values. The project does not claim that pointwise integral identities alone prove completeness or that the entire L^2 spectrum has been formalized.

Reproduction and references

Lean 4.19.0 is pinned, with Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

Run `python verify.py` at the root for supplementary exact algebra checks. Compile with `tectonic main.tex`. Actual fresh-build logs and standard-foundation axiom reports are included. Author, parent and internal cross-review were performed; no external independent review is claimed.

References

[1] The Last Math Competition, Conjecture 00000009778. Exact bilingual source included.

[2] M. R. Opmeer, *Decay of singular values for infinite-dimensional systems with Gevrey regularity*, Systems & Control Letters 137 (2020), 104644, Example 31. Publisher record; author manuscript, PDF page 23.